
What Sets the Ceiling on Automated Code Audit: the Auditor, its Rulebook, or the Specification?

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Abstract

An automated auditor reads code it did not write and decides whether to block it. We ask what sets the limit on how much it catches, against ground truth no model supplies: hidden test suites executed by an interpreter. Holding the auditor, its instructions and its ground truth fixed, and varying one thing at a time, we find the limit is set jointly by which model reads, what its rulebook tells it to look for, and whether the benchmark’s hidden suite asks a question the specification answers.

Returns to repeated reading diminish well short of complete. Eight independent readings of the shipped cross-vendor auditor reach 30.0% union recall on defective increments [20.0, 40.7], at 16.0% on correct ones [10.1, 22.3]; the eighth reading still adds 1.9 points [1.1, 2.8], above the 1.0-point flattening bar registered in advance. We therefore report diminishing returns at the budget we could afford and do not claim saturation. Twenty readings pooled over three auditor families reach 48.2% [36.7, 60.0], though those twenty are not twenty opportunities: one of the three routes gains 1.8 points from its first reading to its eighth against another’s 19.3.

Which model reads moves the limit further than how often it reads, though not in a way that survives its own controls. A stronger same-vendor model scores -26.4 points of union recall $[-37.3, -15.6]$, a preregistered primary, while flagging correct code at 3.3% against 16.0%. Under a weaker severity rule the sign reverses, and the two routes are not sampled alike: one sends temperature 0 and the other sends no sampling parameter at all, so their unions are not equally free to grow. We read the contrast as evidence about severity calibration and sampling rather than about auditing ability, and we report a third

model that beats both on recall and on false positives at half the readings. Adding one sentence to the rulebook, telling the auditor to look for what the visible tests do not cover, moves flags by $+26.8$ points [13.6, 40.4] on an exploratory 56-instance arm, at 12.5 points more flags on correct code whose admissible interval $[-3.6, +26.8]$ does not exclude zero; on the contrast that isolates the sentence, hidden-suite pass rate moves $+5.4$ points $[-0.9, 12.1]$ and does not exclude zero.

Much of what the auditor misses is failure the specification never determined. Under a rule asking whether the prose settles the hidden suite’s expected value, two raters — a model and an author of this paper, who wrote the rubric and was not blind to the earlier labels — agree that 44 of the 57 never-flagged defects are cases where it does not, reversing the same instances’ earlier classification and withdrawing a claim this work previously made. Where an outside party had independently flagged one of these specifications as defective, our raters agree on 9 of the 11 overlapping tasks.

Two cautions follow for work that reports critic and reviewer recall. A union-recall figure is contingent on the matching rate on correct work, so recall quoted without it is not portable across substrates or auditors. And repeated reading buys coverage only in proportion to how much of the population sits near the auditor’s decision boundary: an auditor sampled deterministically returns one verdict however often it is asked, and its union over K readings is flat by construction.

Every study was preregistered before its first model call, with hypotheses, intervals, kill conditions and stopping rules fixed in advance, with one exception this paper names, and four preregistered hypotheses were killed by their own rules. Reports that have passed an independent review by a model from a different vendor with access to the raw records are marked as such. The four studies that were in review when this paper was first written have since passed, after between four and

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eleven rounds each; two more have since passed at seven rounds and six; and one, an intervention that adds the missing behavioural rule to the specification and re-audits the same code, is in its first round and licenses nothing here until it settles. Every number this paper draws from a report under review is labelled where it appears. The reviews are archived verbatim beside the results they judge.

1. Introduction

An automated auditor reads code it did not write and says whether to block it. We ask how much such an auditor catches, and what decides the limit.

That question decomposes. A reviewer's score on a benchmark mixes at least four things together: whether the model can see a defect, whether it judges the defect worth blocking on, whether the benchmark's hidden tests define the behaviour it would have to see, and how many times it is allowed to look. Work on code-review and critic models improves the first two and reports the aggregate gain. We hold the auditor, its instructions and its ground truth fixed, and vary the other three one at a time, against a truth no model supplies: a hidden test suite executed by an interpreter.

Returns to reading again diminish. Eight independent readings of the same increment by the shipped cross-vendor auditor reach 30.0% union recall on defective increments, with a 95% problem-cluster interval of 20.0 to 40.7, at 16.0% [10.1, 22.3] on correct ones. The eighth reading still adds 1.9 points [1.1, 2.8], above the 1.0-point flattening bar we registered in advance, so the curve had not flattened at the budget we could afford and we do not claim saturation. Twenty readings pooled across three auditor families reach 48.2%.

Which model reads, and what it is told to look for, move the limit further. Swapping in a stronger model of the generator's own vendor changes union recall by -26.4 points [-37.3, -15.6], a preregistered primary. That contrast does not survive its own controls: the stronger model flags correct code at 3.3% against 16.0%, the sign reverses under a weaker severity rule, and the two routes are sampled differently, one at temperature 0 and one with no sampling parameter sent, so their unions are not equally free to grow. A third model beats both on recall and false positives at half the readings. Adding one sentence to the auditor's rulebook, telling it to look for what the visible tests do not cover, moves flags by +26.8 points [13.6, 40.4] on a 56-instance exploratory arm, at 12.5 points more flags on correct code — a cost whose admissible interval is [-3.6, +26.8] and does not exclude zero, so it is the pooled contrast rather than the correct stratum alone that puts the rule outside the product's constraint;

on the contrast that isolates the sentence, hidden-suite pass rate moves +5.4 points [-0.9, 12.1] and does not exclude zero.

Much of what the auditor misses turns out to be failure the specification never determined. We re-rated the never-flagged residual under a rule that asks whether the prose settles the hidden suite's expected value before it asks whether the visible tests exercised the input. Two raters agree that 44 of 57 residual instances are cases where the prose does not settle it, reversing the same instances' earlier classification. Where an outside party had already flagged one of these specifications as defective, for their own purposes and before we looked, our raters agree on 9 of the 11 tasks that overlap.

We tried to test that prospectively, by constructing a population whose specification-determinedness would hold by construction, and the attempt failed in a way worth reporting. Independent review established that our construction never enforced the property: the filter registered to recover the failing input is a truthiness check on a field the injector wrote — of 281 recorded witnesses, 179 are strings and 102 are lists, and none is executed — and four of the other five filters accept cases their registered wording excludes. What the population actually contains is small injected edits that survive a sparse visible suite and fail a hidden one. A preregistered probe separates those edits from natural code at 96.7% on the items it answered, *by a feature it does not identify*: its two arms share no problem, so task composition alone could produce the result. The prospective test therefore does not exist yet, and the classification above remains post hoc.

A union recall figure is contingent on the false-positive rate it cost, so the ceiling we measure describes an operating point rather than a constant. We argue that as a methodological point and not from a second measurement: we attempted one on a substrate with much longer specifications, and it closed without a quotable number after three runs (§3.5). Its clearest lesson is the one we do not get to quote — across three extractions of the same protocol, the quantity we had most wanted to use, whether the two substrates' false-positive intervals meet, came out apart, overlapping, and apart again.

1.1. Preregistration and review

Every study was preregistered before its first model call, with hypotheses, intervals, kill conditions and stopping rules fixed in advance, and every departure recorded as a numbered amendment that says what was known at the time, with one exception the paper names and whose registration is withdrawn. Four preregistered hypotheses were killed by their own rules, two of them our own conjectures, and one of those kills requires a headline in this paper to be stated

as a flag rate rather than a recall. Reports are reviewed by a model from a different vendor with access to the raw records, several needing many rounds, and the reviews are archived verbatim beside the results they judge. Every study reported here has completed that review, at between four and eleven rounds each, and Section 3.5 collects the six that were still in it when this paper was first written. A seventh, an intervention on the specification, has been refused twice and licenses nothing here. We record this as method, and it is what lets a reader read the negative results here as measurements.

2. Method

2.1. Substrates and ground truth

Both substrates are Python functions with executable hidden test suites, so every stratum, recall denominator and false-positive population is decided by an interpreter and no model judges any outcome. **Substrate 1** is HumanEval+ and MBPP+ (EvalPlus), whose median task is 41 words of prose and 6 lines of reference code, with the benchmark’s base suite shown to the generator and its plus suite held back. **Substrate 2** is the standard-library half of BigCodeBench, whose median task is 118 words and 41 lines. BigCodeBench ships one test class per task, so the visible and hidden suites are separated here: two test methods chosen by a fixed seed are the visible suite and all methods are the hidden suite, with the class run unmodified and told which methods to execute. Tasks whose declared modules are unavailable in the frozen interpreter, or whose reference solution fails its own suite there, are dropped before generation and counted.

Candidates are written by one generator shown the specification and the visible tests. A candidate that passes the visible suite and fails the hidden one is a **defective** instance; one that passes both we call **correct**; one that fails the visible suite is discarded. Recall is measured on the defective stratum and false positives on a sample of the correct stratum.

Two things follow from those definitions and are meant literally throughout. *Correct* means *passes the hidden suite*: it is a benchmark label produced by an interpreter, not a guarantee that the program meets its specification, and a suite can be incomplete or wrong. Since this paper’s own third result is that a specification often fails to determine what the hidden suite asks, we cannot also treat passing that suite as correctness in the ordinary sense, and a *false positive* is likewise a flag on test-passing code rather than a demonstrated mistake by the auditor. And the defect population is not a sample of model-written code: it is the **residue that already survived the visible tests**, which is by construction the harder part. A recall figure here is the fraction of that residue an auditor catches, never the fraction

of all defects a model writes.

2.2. The auditor and the estimand

The auditor reads the specification, the visible tests and the candidate, and returns findings with severities; an increment is blocked when any finding is graded BLOCKER. It never sees the hidden suite. The estimand throughout is **union recall at K** : the fraction of defective instances blocked by at least one of K independent readings, averaged exactly over all $\binom{K_{\max}}{K}$ subsets rather than by sampling. The matching quantity on the correct stratum is the union false-positive rate. A union recall figure is reported only beside the false-positive rate it cost.

Four auditor routes appear below. The **shipped cross-vendor** auditor is gpt-5.6-terra; the **same-vendor** auditor is claude-haiku-4-5, the generator’s own family and size; the **stronger same-vendor** auditor is claude-sonnet-4.6; and a third family, gpt-6-astra at high reasoning, is read at $K = 4$. Candidates on both substrates are written by claude-haiku-4-5. Prompts, the auditor’s constitution and every capability card are in the repository.

The routes are **not sampled alike**, and this bounds every cross-family union-of- K contrast in the paper. The cross-vendor route’s capability card does not permit a sampling temperature, so no sampling parameter is sent and the route varies between readings; the same-vendor routes permit one and the layer sends temperature 0, so those routes are near-deterministic. A union over readings that do not differ cannot grow, so the two families’ unions are not equally free to rise with K , and a cross-family difference in union recall confounds auditing ability with how much each union was permitted to grow. The asymmetry is a property of the shipped routing layer we measured rather than a design choice, we did not discover it until after the runs, and we report the $K = 1$ contrast beside the $K = 8$ contrast wherever it matters. A temperature-matched replication is the first item in §4.5.

2.3. Statistics

The unit of observation is the instance and the resampling unit is the problem, because problems recur across generation batches and instances sharing a problem are not independent. The K readings of an instance are repeated measurements of it, not K further independent observations. The bootstrap over problems estimates uncertainty from which tasks the benchmark happens to contain; it does not estimate how much a fresh batch of model calls on these same tasks would move the numbers, which would need replicate runs we did not budget for at every arm. Every rate carries a 95% percentile bootstrap over whole problem clusters (10,000 resamples, seeds fixed in each preregistration),

with the Wilson interval printed beside it and marked as too narrow. Paired contrasts carry an exact McNemar test and a cluster-level sign-flip permutation test; where every discordant pair points one way the percentile bootstrap's bound at zero is an artefact and an exact unconditional interval is quoted instead. A saturation curve is summarised by a constrained exponential fit, and the fitted asymptote is quoted only when the preregistered flattening bar is met; otherwise it is labelled an extrapolation and the raw union at $K_{\max}$ is the number quoted. Counts of five or fewer are quoted as counts. Multiplicity is handled as a single Bonferroni family per study over every comparison actually computed, not over those planned: in the study carrying the primary contrasts that family has sixteen members and a threshold of 0.00313, and two comparisons clear it. **Interval coverage is measured rather than assumed or waved at.** Under the two strata's actual cluster profiles — 110 instances in 56 problems, and 150 in 143 — with within-problem concordance 0.5, the percentile bootstrap's coverage of a nominal 95% rate interval is:

true rate	0.03	0.05	0.10	0.16
defect stratum (56 clusters)	0.797	0.870	0.932	0.932
correct stratum (143 clusters)	0.887	0.917	0.938	0.932

Coverage is near nominal at the rates this paper's headlines sit at and **badly anticonservative at small rates**: an interval quoted around 3% on the defect stratum covers about 0.80, not 0.95. Intervals on rates below roughly 0.10 should be read as narrower than they claim. *An earlier version of this paper said the bootstrap under-covers "by roughly 2 to 5 points" and that its coverage under the clustered design was unvalidated. The first was a generalisation the source report had already withdrawn as unsupported; the second was wrong, since a clustered simulation sat in this paper's own test suite. A pre-submission referee found both.* The simulation is `analysis/coverage.simulation.py` and costs no model calls. Two earlier paired-contrast constructions in this programme were found by review to cover 0.416 and 0.075 where 0.95 was claimed, and both were replaced. No sample size was justified in advance by a minimum-detectable-effect calculation.

An interval that contains zero is reported as one that does not exclude zero, and nothing more. No equivalence margin was registered anywhere in this programme and no equivalence test was performed, so none of those intervals is evidence that two conditions are alike or that an intervention did nothing — a distinction this work has already had to make the hard way, having once inferred equal cost from an interval containing zero and withdrawn it on review. Where a hypothesis fails its registered criterion we say it failed, not that the effect is absent.

All analysis is plain Python 3.13 using only the standard library — `random`, `math`, `itertools`, `csv` and `json`. The bootstrap, the exact McNemar test, the cluster sign-flip permutation and the exact subset enumeration behind every union-at- K figure are implemented in the repository rather than called from a numerical package, so each is readable at the point where it is used, and none depends on a third-party version.

2.4. Preregistration and independent review

Each study was preregistered before its first model call, fixing its population, hypotheses, primary outcome, intervals, kill conditions, budget and stopping rule — **with one exception this paper reports rather than elides**. The analysis behind §3.5's third-rating result was registered a day *after* its contrast had been computed and committed, in a document that claimed the opposite; that document is withdrawn in the repository and the analysis is registered nowhere. Its method was fixed in advance by an earlier amendment, and the disjoint reading, the abstention-excluded secondary and the broken-sheet comparison were not. Departures are numbered amendments committed before the step they govern, each recording what was known at the time; where an amendment followed an interim observation, the results say so and argue whether the change could favour the hypothesis.

Each report was then reviewed by a model from a different vendor to the generator, with read access to the raw records, asked to re-run the analysis, recompute the intervals independently, and check every sentence against the records. **That reviewer is gpt-6-ast-ra, which is also one of the auditor families measured in §3.2 and one of the two raters behind §3.4's labels.** For most claims this is a cross-vendor check on work the reviewer had no part in. **For the claims that rank ast-ra against the other families it is not: there the reviewer is the subject**, and those comparisons should be read as self-assessed until a model that audited nothing in this programme has checked them. This project's own P3 Amendment 10 disqualified the same model as a rater for the same reason, and the standard was not applied here until a pre-submission referee pointed it out. Reviews are archived verbatim, including those that refused approval, and several studies needed many rounds: on one day five reports were refused at once, and clearing those five took thirty-one further rounds. *No measurement moved in any of them.* What moved was what the reports claimed about their measurements — and in most of the later refusals the object of the refusal was a repair written to satisfy the previous round, not the study it repaired. We state that because it is the most transferable thing this process produced: a fix written to satisfy a reviewer is itself an unreviewed claim.

The reviewer is a language model, and this paper is about the ceiling on language models reading code they did not

write. That applies to our own instrument, and the round counts are the evidence: the studies here cleared review at rounds **4, 6, 7, 9, 10 and 11**, and in every case the round before the last still returned substantive findings. A single pass with high recall would clear a report in one round or two. So *quotable at round N* means a report survived N passes, not that it was verified, and the marks in this paper should be read as a filter of unknown recall rather than as proof.

Two things sharpen that. First, the split: across the six studies of §3.5, *no measurement moved*, though the scope matters and two counterexamples sit just outside it. The clarification study’s second review moved its primary from 9 of 31 to 9 of 32 instances and its contrast from +29.0 to +28.1 points, by finding an instance wrongly excluded; and a label count inside a reviewed report was 71 where its data say 67. Both are measurements that moved on review, and an earlier version of this sentence said none ever had. The reviewers reproduced rates, intervals, subset enumerations and seeds independently and found no arithmetic error, and every refusal was about what the prose claimed of arithmetic already confirmed. On recomputation the instrument is strong; on interpretation it is weak, and the two should not be quoted as one assurance. Second, the class of defect this leaves open is the one that looks right to a language model. The clarification study audited the wrong program for 384 readings — a per-problem field used where a per-instance one was needed — and every automated gate passed, because each checked that the code was *identical across conditions* and it was, identically wrong. It was found by hand, reading an adjudication sheet and noticing that a displayed program returned exactly what its hidden suite was said to reject. Neither the gates nor a reviewer found it, and we do not know what else of that kind remains.

Every review prompt in this programme stated the claimed result before the reviewer began. We attempted to test what that costs, and the attempt failed in a way worth reporting. One report, quotable at round 6 after five refusals and unchanged since, was dispatched again under a prompt whose own text withheld every figure and every prior verdict and asked for an independent computation first. **That prompt’s list of binding artefacts then named two files that state those figures**, one of them the claims-ledger entry for the result under review. The blinding was not enforced, so the agreement between that pass and the briefed one establishes nothing, and we draw no inference from it. What the pass did return is a *disagreement count*: six substantive findings that six briefed rounds on the same file had not produced, among them that the report’s two groups were two different definitions of *caught* rather than one population with duplicate rows, that its rating sheet displays at most three failing inputs while 81 of its 121 entries have more recorded, and that a count of changed labels was 71

where the data say 67. ****That count is $n = 1$, uncontrolled and unregistered****: a seventh round might have found new things under any prompt, and no matched briefed round was run against the same file. We report it as a reason to test blinding properly rather than as a result about blinding, and the first round of every review after this one is dispatched with the conclusions withheld from the prompt *and* from every artefact it names.

Every study drawn on here has completed that review. One further study was re-run twice, closed without a quotable number, and appears nowhere in this paper except in this sentence.

3. Results

The claims this paper defends, each with its interval and its label, are listed in `CLAIMS.md`. A claim enters that file only after its report has passed an independent review by a model from a different vendor with access to the raw records. Every section below draws only on reports that have passed such a review. Section 3.5 reports four studies whose reviews took between four and eleven rounds each to reach that point, and states what each of them does and does not license; none of them is load-bearing for §3.1–§3.4. Rates carry the 95% problem-cluster percentile bootstrap as the primary interval, with the Wilson interval beside it where the source report gives one, because problems recur across generation batches and an interval that treats instances as independent is too narrow. The coverage of that bootstrap under this clustered design with 56 problem clusters has not been validated by simulation, and every primary interval in this paper inherits that caveat.

Sixteen comparisons were computed in the study that carries the primary contrasts, under one Bonferroni family with a threshold of 0.00313. Where a p value appears below, the exact McNemar value and the cluster-level sign-flip permutation value are both given, because instances within a problem are not independent and only the second respects that.

3.1. Returns to repeated reading diminish short of complete

The shipped cross-vendor auditor reads an increment holistically and returns findings; an increment is blocked if any finding is graded BLOCKER. Over eight independent readings of the same 110 defective instances, union recall rises from 10.7% at one reading to 30.0% [20.0, 40.7] at eight (Figure 1). The eighth reading still adds 1.93 points, cluster [1.14, 2.78], against the flattening bar of 1.0 points registered before the first model call. The bar was not met. The constrained fit puts the asymptote at 31.5%, but on our own registered criterion that figure is an extrapolation and the

raw union at eight readings is the number to quote. What the data support is diminishing returns at the budget we could afford, not saturation: at 1.9 points per reading the curve was still climbing when we stopped. That bar is also a function of where a family was stopped, so families run only to $K = 4$ meet it more easily than families run to $K = 8$, and a curve that meets it at a small $K_{\max}$ is making a weaker statement than one that meets it at a large one. Twenty readings pooled across three auditor families reach 48. **Those twenty readings are not twenty opportunities.** The pool is eight cross-vendor, eight same-vendor and four *astra* readings, and the same-vendor route gains **1.8 points** from its first reading to its eighth against the cross-vendor route's 19.3, because it is sent temperature 0 and returns nearly the same verdict each time. The effective depth is closer to thirteen, and the pooled figure should be read as a union over three routes rather than over twenty independent looks.2% [36.7, 60.0].

The price is measured on the same footing. Union false positives on 150 correct increments reach 16.0% [10.1, 22.3] at eight readings. The recall bought per point of false positive — recall gained over one reading divided by false positives gained over one reading — is **1.68** at eight, and it does not worsen as readings accumulate: the ladder is 1.67, 1.63, 1.64, 1.65, 1.66, 1.67, 1.68 from $K = 2$ to $K = 8$. An earlier version of this paper said it falls monotonically; the source table contradicts that — the ratio falls once and then rises, so repeated reading on this route buys recall at a slightly improving price, not a worsening one. It is a ratio of two measured rates and carries no interval of its own.

3.2. Which model reads, and under which severity rule

Replacing the auditor with a stronger model of the generator's own vendor moved union recall at eight readings by -26.4 points, cluster $[-37.3, -15.6]$: 4 of 110 against 33 of 110. Three instances were flagged only by the new family and thirty-two only by the shipped auditor, at an exact McNemar p of 4.2×10^{-7} and a cluster sign-flip p of 2.0×10^{-5} . This was a preregistered primary. Three things qualify what it means, all of them measured here and none of them favourable to reading it as a statement about auditing ability.

First, the two auditors are not at the same operating point, which is the comparison this paper elsewhere warns against. The stronger model flags correct code at 3.3% — 5 of 150, a count the statistics section says to quote as a count — whose bootstrap interval [0.7, 6.7] sits in the range where that interval covers about 0.89 rather than 0.95, so it is narrower than it claims. Against the shipped auditor's 16.0% [10.1, 22.3]. It blocks less of everything.

Second, the sign reverses under a weaker severity rule (Figure 2). Counting any finding rather than only findings

graded BLOCKER — an exploratory rule, not preregistered — the stronger model reaches 59.1% [47.3, 70.6] on the defect population at 34.0% [26.5, 41.7] on correct code, against the shipped auditor's 34.5% [23.6, 45.5] at 19.3% [12.8, 26.3]. The ratio of those two rates — a derived point ratio we carry no interval for — is 1.74 and 1.79 respectively, so nothing here shows one family buying more recall per false positive than the other. The stronger model writes down more of what is wrong and grades less of it as blocking. The -26.4 is a fact about severity calibration.

Third, the two families are not sampled alike. The shipped cross-vendor route's capability card does not permit a temperature, so no sampling parameter is sent and the route varies between readings; the same-vendor route permits one and the layer sends temperature 0, so that route is near-deterministic. A union over near-deterministic readings cannot grow, and the two unions are therefore not equally free to rise. They do not: over $K = 1$ to 8 the cross-vendor family gains 19.3 points and the same-vendor family gains 3.0. At one reading the gap between them is -10.1 points, not -26.4 ; roughly two-thirds of the headline is the difference in how much each union was permitted to grow. We regard this as the most serious limitation of the contrast, and §4.5 records what would be needed to remove it.

A third family argues the other way and belongs beside the headline. A high-reasoning model of the cross-vendor family reaches 32.7% [20.7, 45.0] union recall at four readings while flagging 10.7% [5.9, 16.1] of correct code, dominating the shipped auditor on both axes at half the readings. So the evidence does not support "a stronger model is a worse auditor". It supports the narrower statement that this particular same-vendor model, at this severity calibration and sampled at temperature 0, blocks far less.

A same-vendor auditor of the generator's own size was worse than the cross-vendor one on these instances by -12.7 points $[-25.0, -0.9]$ of union recall at eight readings, signed so that a negative value favours the cross-vendor auditor. We reported until 2026-09-21 that a study on a second substrate reverses that sign. *That is withdrawn*: the study closed without a quotable number, and on its clean run the same contrast included zero, so no reversal was established either (§3.5). The figure above is therefore a measurement on this substrate, and whether it is a property of same-vendor auditing is open — which is what it was before we attempted the second substrate, not something that study settled in either direction.

3.3. One rulebook sentence moves flags, and does not measurably move outcomes

Adding a single rule to the auditor's constitution, telling it to find what the visible tests do not cover, moved flags on the defect population from 10 of 56 to 25 of 56: $+26.8$ points,

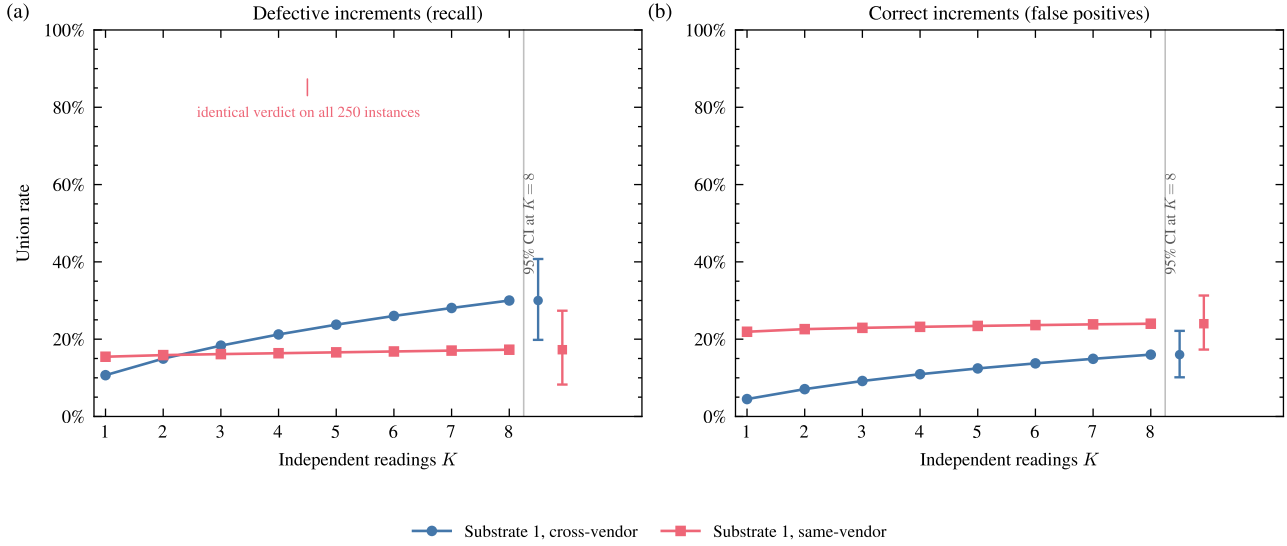


Figure 1. Returns to repeated reading diminish short of complete. Union rate against the number of independent readings K , for two auditor families on substrate 1. (a) recall on defective increments; (b) false positives on correct increments. Bars at the right of each panel give the 95% problem-cluster bootstrap interval of each series at $K = 8$, dodged in x so overlapping intervals stay separable. Neither curve meets the preregistered flattening bar: the cross-vendor last-step gain is 1.93 points [1.14, 2.78] against a bar of 1.0, so neither is described as saturating. The same-vendor route is sent temperature 0 and the cross-vendor route no sampling parameter at all, so the two unions are not equally free to rise with K (§3.2). A second substrate was plotted here until 2026-09-15; it was re-run twice and closed without a quotable number (§3.5).

cluster [13.6, 40.4], exact McNemar $p = 0.0003$, cluster sign-flip $p = 0.0009$. With the pooled defect-and-correct flag contrast this is one of only two comparisons that clear the Bonferroni threshold, and the two are the same effect seen twice.

The estimand is narrower than the sections above and must be read at its own scale. It is measured on 56 instances from 41 problems, not the 110-instance defect population; it counts flags, not union recall against the hidden suite; it is one reading, because every loop arm revises once, not a union over eight; and it is exploratory, because the preregistration named outcome contrasts rather than flag contrasts. For those reasons we do not compare its magnitude with the -26.4 or the $+19.3$ of the preceding sections, which are union recall at $K = 8$ over 110 instances. They are different quantities on different populations.

Two costs travel with it, and the first is weaker than an earlier version of this paper said. On correct code the same rule moved flags by $+12.50$ points. ****Every discordant pair points the same way**** (7 against 0), so by the rule stated in §2.3 the percentile bootstrap’s bound at zero is an artefact and the admissible interval is the exact unconditional one, which is $[-3.6, +26.8]$ and **does not exclude zero**; its exact McNemar p is 0.0156 against that study’s Bonferroni threshold of 0.003125. Earlier versions of this paper quoted the forbidden bootstrap interval [5.17, 21.82] in four places, including the abstract. The directed statement

the records support is the pooled one, where 23 discordant pairs point one way and 1 the other at $p = 3.0 \times 10^{-6}$: the sentence makes the auditor louder overall. On the correct stratum alone the cost is $+12.50$ points with an interval that contains zero. And on the contrast that actually isolates the rule — the referent arm against the cross-vendor arm on hidden-suite pass rate after one revision — the effect is $+5.36$ points, cluster $[-0.89, +12.07]$, $p = 0.146$, which does not exclude zero. The rule demonstrably moves what the auditor writes down. Whether that converts into code that passes the hidden suite is not established, and since this paper argues elsewhere that a flag is not a catch, that distinction applies to our own intervention first.

3.4. Most of what the auditor missed is failure the specification never determined

Fifty-seven of the 110 defective instances were flagged by no reading of the three families read at that point. That is a statement about twenty readings, not about detectability: if an instance is caught with probability p on each independent reading, twenty readings miss it with probability $(1 - p)^{20}$, which is still 98% at $p = 0.001$ and 36% at $p = 0.05$. Never flagged here therefore does not mean never flaggable, and we use the residual as a population to characterise rather than as a set of defects shown to be invisible.

It is not never flaggable, and we can put a number on that. Reading six families at forty readings, includ-

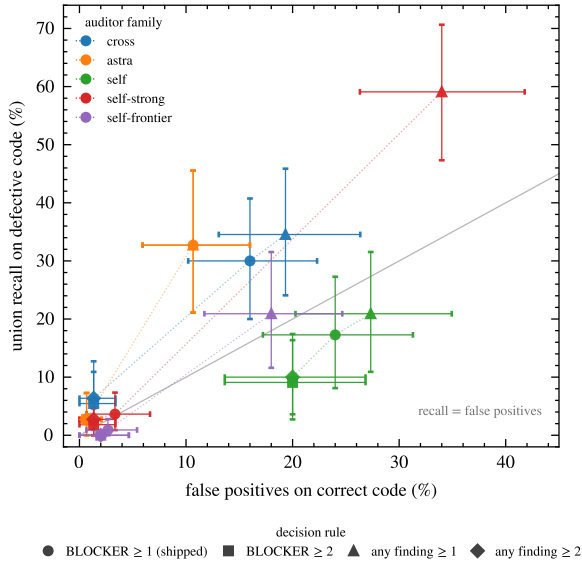


Figure 2. Flag coverage is a property of the decision rule, not only of the auditor. Each point is one auditor family re-graded under one of four decision rules on the same archived readings, with problem-cluster 95% intervals on both axes; no new model calls were made. The dotted guides join a family’s four points and are *not* a measured curve — only the four points exist. Two things the accompanying table cannot show: the families do not lie on one frontier, and *self-strong* has no reachable point between 3.3% and 34.0% false positives, which is exactly where a comparison matched to the cross-vendor auditor would sit, so its ranking there is bracketed rather than measured. Nothing in this figure assesses finding *content*; a flag is not a diagnosis.

ing the same auditor under a rulebook with one sentence added, shrinks the never-flagged set from 57 to **32 of 110**, and **43 of the 68 instances both raters call specification-undetermined are flagged by some reading of some family**. Being undetermined is not sufficient for being missed. What does not move is the composition: the undetermined share of the never-flagged set is **44 of 57**, 77.2%, at three families and twenty readings, and **25 of 32**, 78.1%, at six families and forty readings. The residual loses nearly half its members and keeps its shape. *An earlier version of this section reported only the 57 and described it as flagged by no reading of any family, which was true of three families and false of the six this programme measured; a pre-submission referee found it.*

The first classification of that residual asked whether the visible tests exercised the failing input before it asked anything else. It called 46 of 57 unexercised edges, and an earlier draft of this paper rested a claim on that.

Re-rating the same instances under a rule that asks first whether the specification’s prose determines the hidden suite’s expected value reverses it. Two raters agree that 44 of 57 (77.2%, cluster [62.1, 91.1]) are cases where the prose

does not settle the value, and that 2 of 57 are cases where it does and the visible tests merely never built the input. The preregistered condition for withdrawing “the residual is mostly unexercised edge” fires on both of its clauses, and the claim is withdrawn. The interval is conditional on a category and has no committed coverage simulation, so it is uncalibrated, as is every category-conditioned interval we report.

A classification that reverses when the order of its questions changes is, by itself, weak evidence about the auditor, and the raters are not independent of it: one was the author, who wrote the rubric and knew the earlier labels and which claim was at stake. Three things support the second reading. The rubric was fixed before any label was written. The labels were committed before we looked for outside evidence. And when a first pass was found to have carried instance identifiers and a sheet name into the raters’ view, the labelling was re-run with identities stripped and the result held. Where an outside party had independently recorded that one of these specifications is defective, in twelve MBPP tasks named in a paper written for another purpose, our raters agree on 9 of the 11 that fall inside our defect population, with both disagreements on the boundary that paper’s own “incomplete” category draws. That check is one-sided by construction: it can support agreement where an outsider looked, and can bound nothing about the rate.

3.5. Six studies, and what they license after between four and eleven review rounds each

Four further studies had reported when this paper was first written, and all four were refused quotation by their independent reviews on the same day. They have since passed, after between four and eleven rounds each. Two more have since reported and passed, at seven rounds and six. We report them here rather than folding them into the sections above because three of the four are narrow or negative results whose limits are part of the claim, and because the shape of the refusals is itself a finding about this method.

A rulebook sentence moves a second vendor’s family. The referent rule — *find what the visible tests do not exercise* — raises union BLOCKER recall on *self-strong* (Sonnet 4.6) at four readings by **+20.9 points**, cluster [+10.8, +31.8], at a false-positive cost on correct code of **+8.0 points** [+3.4, +12.7]. The grading rule shows no demonstrated improvement: +0.9 points [−1.8, +4.5], and that hypothesis fails its registered criterion, which demonstrates neither equivalence nor that the rule did nothing. This reproduces §3.3’s effect on a different vendor’s model at a different K , with its own price. A post-hoc quantity from the same study — whether a flagged finding *reports a failure* on the named class — is **3 of 110** under the shipped constitution and **27 of 110** under the rule, and *may not be quoted as defect recognition*:

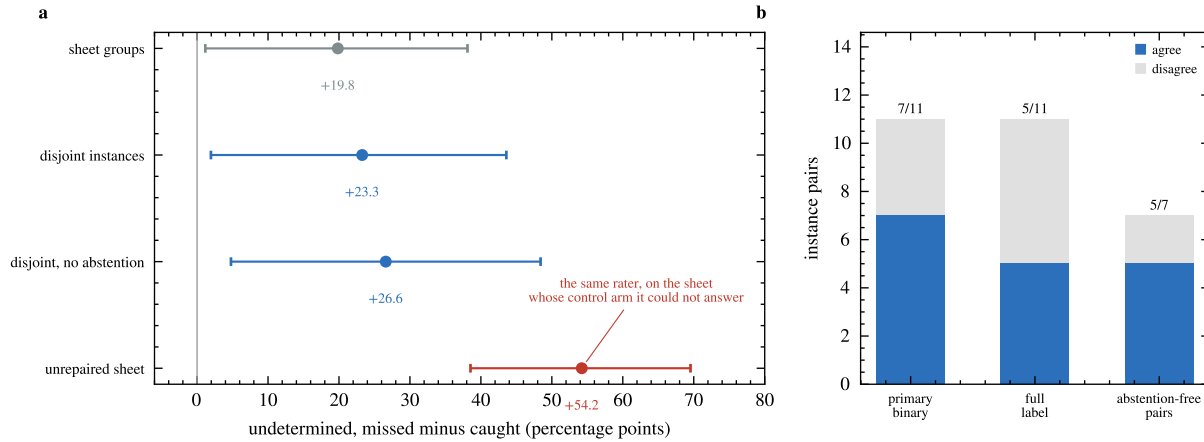


Figure 3. The residual contrast, printed beside the instrument that produced it. a. The share of instances a rater judged specification-undetermined, missed group minus caught group, under each reading, with problem-cluster 95% intervals. The two sheet groups use different definitions of *caught* and share eleven instances, so the disjoint readings are the ones that compare populations; both are *post hoc*. The red point is the same rater on the same sheet before its control arm was rebuilt, when 42 of 53 caught entries carried no failing-input evidence and all 42 drew *cannot-tell*: an unanswerable control arm did not produce noise, it produced a larger contrast pointing the way the hypothesis points, and a preregistered abstention gate is what excluded it. **b.** The same rater’s agreement with itself on the eleven instances the sheet carries twice, with identical text and identical evidence. **The contrast in a is read off the instrument in b,** and the claim this paper draws from it is stated under that constraint.

its rubric admits a finding even where it grades the class non-blocking or says the specification is silent, so findings that disclaim the requirement are counted in. Seven review rounds passed over the opposite error, in which the report called that instrument *narrower* than defect assertion — the direction that makes a quantity look more conservative than it is.

A measured 60.9% is not a higher ceiling, and it is the same estimator as the 30.0%. On the defect population the same rule takes union recall — the identical quantity, blocked by at least one BLOCKER, unioned over eight readings — from 33 of 110 to **67 of 110 = 60.9%** [48.6, 72.5], **+30.9 points** [19.1, 43.1], at **+20.7 points** [12.8, 28.8] of false positives, with a single-draw rate of 23.6% [17.1, 30.3] that fails the product bar on all eight draws. *Earlier versions of this paper called 33 of 110 union recall and 67 of 110 flag coverage, and a pre-submission referee observed that the name tracked which number suited the framing.* One name is used here for one estimator, and The caution that a flag is not a catch attaches to both equally. What separates them is not their status as measurements but their price: 36.7% [28.5, 44.8] false positives against 16.0%, which is why the shipped route is the shipped route. The fitted asymptotes differ by +27.1 points [10.9, 39.0] and *that does not establish a higher ceiling*: the cross-vendor arm never flattened, so one of the two terms is an extrapolation; heterogeneous low-probability detection imitates a ceiling at this budget, since an instance found with probability 0.02 per reading survives eight readings about 85% of the time; and this study’s own zero-inflated estimator finds no never-detectable class in

either arm. Its adjudication kill is arithmetic rather than a labelling artefact — only 9 of 110 instances drew any finding, so the largest defect-asserting count available under perfect adjudication was 9, below the floor of 20. Every adjudication-derived figure in it is conditional on labels whose production can no longer be audited: one rater’s raw reply, prompt, launcher and execution log lived only in a session scratchpad and were destroyed between the review that flagged them and the repair.

Corroboration does not validate a generated test. A wrong generated test — one that fails on the dataset’s canonical solution — is not reliably recognised by asking whether further independent generation draws also fail on the same candidate. Neither paid rule reaches the preregistered threshold: retained wrong-test rates are **12.2%** (Wilson 7.0–20.6; cluster 3.6–22.6) and **14.9%** against a limit of **2%**, and neither beats the free within-draw comparator at 12.8%. A floor of 2.4% holds on the original unique-test quantity because 29 of the 45 wrong tests never fail on any candidate at all. The claim is negative and substrate-bound: these rules do not validate generated tests here, not that generated tests cannot be validated.

An injected population does not give the residual result a prospective test. We constructed a population of injected defects intended to make specification-determinedness hold by construction. It does not. The filter registered to recover each edit’s first failing hidden input is a truthiness check on a field the injector wrote; of 281 recorded witnesses 179 are strings and 102 are lists, none is executed, and four of the remaining five filters accept cases their registered

wording excludes — a suite that reports success after an early exit, a process that terminates without an assertion or an exception, an added import from a module already imported. *No filter establishes specification entailment, and repairing the first one alone would not have*: recovering the failing input and executing a witness shows the edit changes behaviour on an input the hidden suite exercises, not that the specification entails the value expected there. What may be quoted is descriptive. At eight readings the auditor blocked **90 of 92** accepted injected instances against **11 of their 92** unmodified twins, 79 discordant pairs one way and none the other, and against **33 of 110** on the natural residual; the instances the construction’s own gate *refused* were blocked **31 of 40**, which is why that gate cannot be assumed conservative. A probe separates the injected set from the natural one at 96.7% on the items it answered, *by a feature it does not identify*: the two arms share no problem, so task composition alone could produce the result, and the probe’s model is one of the two gate models. So §3.4 keeps exactly the status its own study gave it, post hoc and unreplicated. A successor needs an independent party to validate the actual violation *and* the specification-entailing expected behaviour under a fixed, blinded protocol; a second substrate alone does not suffice, and an independent rater alone does not either, because task composition and construction confounding survive both.

A fifth study was re-run twice and closed without a quotable number. A second benchmark was reported here until its review refused quotation and we confirmed the reason: the visible-test text shown to the auditor *and* to the generator was sliced out of its test class without the class header, so it failed to parse on all 300 tasks while scoring executed the intact class. The repair for that fault introduced a second one, re-attaching a hidden method’s decorators to the first visible method on 26 tasks. A third run on a clean extraction reproduced five of the six findings that the second run had reversed and did not reproduce the sixth; the criterion registered before those numbers existed was stated over the set, so the strict reading applies and the study is closed. Three runs, about \$82, and no quotable number. Whether five of six should have licensed the five is recorded as an open question in CLAIMS.md rather than settled by us.

Measured flag coverage is rule-dependent, and a headline’s magnitude belongs to its own rule. Re-grading findings already archived by two of the studies above under four decision rules, with no new model calls, leaves the shipped BLOCKER threshold’s estimate intact for its own rule — which *was* preregistered, as “ ≥ 1 BLOCKER finding” — while *reversing its sign* under any-finding grading. That reversal is a mechanical consequence of changing what counts, and it identifies nothing about why the families produced different findings or different severities. At a

common depth of four readings, *astra* exceeds the cross-vendor auditor on union recall over the defect stratum by **+11.5 points** [+2.9, +21.3], paired within instance. **The two sides are not averaged alike**: the cross-vendor arm has eight readings and is averaged over all $\binom{8}{4} = 70$ four-draw subsets, while *astra* has exactly four, so $\binom{4}{4} = 1$ and its estimate is a single realisation with none of its reading-selection variance averaged out. An earlier version of this paper said 70 subsets on each side; the source report has it right and the manuscript lost it. The contrast should be read knowing that one arm’s luck in four draws is fully in the number and the other’s is not, with a false-positive difference of -0.3 points $[-3.8, +3.4]$. On the cross-vendor auditor’s complete eight-draw ladder the same contrast is $+2.7$ points $[-6.5, +13.0]$ with a false-positive difference of -5.3 points $[-10.3, -0.7]$: that comparison establishes *lower false positives, not higher recall*. Observed false-positive rates near one another are not a matched operating point, and one family has no reachable point between 3.3% and 34.0% false positives — precisely where a matched comparison would sit — so its ranking is *bracketed rather than measured*. Nothing here assesses finding content against the hidden failure, so none of it speaks to what any finding identified.

The ambiguity gap survives a rater who is not an author, and the instrument that shows it disagrees with itself.

The contrast in §3.4 between missed and caught instances had been rated by a model and by an author. A third model re-rated the whole sheet after we rebuilt it on mechanically recovered witnesses — necessary because its control arm had been *unanswerable*: 42 of 53 caught entries carried no failing-input evidence and all 42 drew *cannot-tell*. The share labelled *undetermined* is 43 of 68 against 23 of 53, **+19.8 points** [+1.2, +38.1]; those two groups are not two populations, since 11 instances sit in both arms, and removing the duplicates gives $+23.3$ points $[-2.0, +43.6]$ — *post hoc*, and larger, so the overlap had understated it. The limit is part of the result: on those 11 instances, shown with identical text and identical evidence, **the rater agreed with itself on 7 of 11** on this binary outcome and 5 of 11 on the full label, where the earlier study’s two raters each agreed 11 of 11 (Figure 3). Every interval’s lower end lies within five points of zero, and this remains a model rather than the outside human the design asks for. Separately, the *unrepaired* sheet scored $+54.2$ points $[-38.5, +69.5]$ — 2.7 times the repaired contrast, in the direction that flatters the hypothesis — which is evidence about sheets and not about specifications, and which a preregistered inconclusiveness gate catches.

What the refusals were about. These six studies cleared review at rounds 4, 6, 7, 9, 10 and 11, forty-seven rounds in all; an earlier version said forty-four, which counted four of the studies only from their first refusal. Most of those

rounds ended in a refusal of the *repair* rather than of the measurement. No measurement moved in any of them; every figure above reproduced independently at every round it was checked. The recurring failure was a check or a sentence that claimed more than it delivered: a binding that protected one sentence of a withdrawal spread across six; the same binding rewritten to protect all six, with two of them truncated before the words that carried their meaning; a check that stripped HTML comments and called the result what a reader sees; and a test named for a code path that never called it. A second pattern appeared in the last thirteen rounds and is worth separating from the first: most of those refusals were *overstatements away from our own interest* — a registration whose chronology its own repository contradicted, a withdrawal that overcorrected past what the record supported, a publication prohibition a registered gate never imposed, a limitation credited to another study’s work that the other study had not performed. Modesty is not the safe direction; it is another direction in which a sentence can be wrong. The corrective is not a better patch but a smaller claim, written where the claim was — and a heading, a title and a limitation are claims.

3.6. What follows for a recall number

Taken together, the results above say an audit recall figure is a joint property of the auditor and its material. It moves with which model reads, with what the rulebook tells it to look for, and with whether the benchmark’s hidden suite asks a question the specification answers. It also moves with the severity threshold at which a finding becomes a block, and with how the route is sampled, neither of which is usually reported.

4. Discussion

4.1. What a union-recall number means

The caution this subsection argued — that a union-recall figure is contingent on the rate it costs, so recall quoted without its false-positive rate is not portable — rested on a second substrate that was re-run twice and closed without a quotable number (§3.5). We state the caution as a methodological point rather than as a measured result, because the argument for it does not need our second substrate: recall and false positives move together with any severity threshold, and a critic can be tuned to any recall by blocking more. What our records still support directly is the narrower version of it inside substrate 1, where the same auditor at two severity rules sits at two different operating points (§3.2), and where the shipped auditor’s 30.0% [20.0, 40.7] recall is quoted only beside the 16.0% [10.1, 22.3] it cost.

The clean experiment for the general claim is a severity-threshold sweep within one substrate, which we did not run.

Comparing two substrates was never that experiment: they differ in specification length, solution length, problem population, defect distribution and suite construction at once.

4.2. Repeated reading buys coverage only where the decision is marginal

Union-of- K is the instrument this paper uses to find a ceiling, and it is worth saying plainly what it cannot measure. Part of that needs no experiment at all: a union over readings that do not differ cannot grow, so a union-of- K curve is flat for a route that returns the same verdict however often it is asked. That is arithmetic, not a finding.

The empirical illustration we had for it came from the second substrate, which closed without a quotable number (§3.5), so we no longer offer one. What survives is the design fact behind it, which our own records do establish: the same-vendor routes are sent temperature 0 while the cross-vendor route is sent no sampling parameter at all (§3.2), and over $K = 1$ to 8 the first family’s union gains 3.0 points against the second’s 19.3. How much of that difference is sampling and how much is the operating point, we cannot say, because no temperature-matched arm was run. We report it as a confound on our own contrasts rather than as a result about auditors.

Best-of- K , union-of- K and panel aggregation over critics therefore buy coverage in proportion to how much of the population sits near the critic’s boundary, and only to the extent that the route is sampled at all. An auditor tuned loud enough, or sampled deterministically, is invariant under resampling, and its union over K readings can rise only as far as its marginal cases allow.

4.3. What the ceiling is made of

Repetition is the smallest of the contributions we could measure. Eight readings of one auditor buy 19.3 points of union recall, and twenty readings across three families leave 51.8% [40.0, 63.3] of the defect population untouched — the complement of the pooled 48.2% [36.7, 60.0]. Changing which model reads moves the same quantity — union recall at $K = 8$ over the same 110 instances — by 26.4 points, though §3.2 gives three reasons not to read that as auditing ability. Changing one sentence of the rulebook moves flags by 26.8 points, but on 56 instances at one reading, and we do not place that number on the same scale as the other two.

A fourth contribution is not the auditor’s at all, and we can only describe it post hoc. Under a rule that asks whether the specification’s prose determines the hidden suite’s expected value, 44 of the 57 never-flagged defects fail that test. We attempted a constructed population where the prose does determine the answer; review established that no filter in it enforces that property, and what it licenses is descriptive only

(§3.5). The association between never being flagged and never being determined is partly definitional, because the residual is defined by conditioning on never being flagged, so this is an observation about where the misses sit and not an estimate of how much the benchmark contributes.

The constructed population carries a limitation we measured rather than assumed, and a second one that review established. A frontier model separates its edits from natural ones at 96.7% on the items it answered, between 79.9% and 97.3% once the items it would not classify are counted at either extreme — *by a feature it does not identify*, since the two arms share no problem and the model is one of the two that gated the population. And the construction does not enforce specification-determinedness at all, so it licenses *no* bound on the measured ceiling. We had written that it did. It does not.

4.4. Flagging is not naming

An audit that blocks an increment for the wrong reason has still blocked it, and a recall number counts that as a catch. With the finding texts kept for one reading of the shipped auditor and adjudicated blind by two raters, the auditor returned a finding on 9 of 110 defective instances and asserted the actual failing behaviour on 5 to 7 of them, depending on how the two raters are combined. The raters agreed at $\kappa = 0.391$, which we report beside the rate it produces, because a moderate agreement is part of what the rate is worth. That study’s preregistered kill condition fired on this result, and §3.5 states its headline in the form the kill requires. The figure is a property of one reading and is not a naming rate for the eight-reading union; establishing that would take adjudicating eight readings’ texts, which were not archived.

4.5. Limits

The confound we would fix first is that auditor families are not sampled alike: one route is sent temperature 0 and the other no sampling parameter at all, which bounds how far one union can grow before any difference in ability is measured. Every cross-family union-of-K contrast here inherits it. Second, no auditor is compared with another at a matched false-positive rate; we have two severity thresholds per family and no sweep between them, so the paper cannot say whether the families differ once brought to the same operating point. Third, the primary intervals are problem-cluster percentile bootstraps over 56 clusters whose coverage under this design has not been validated by simulation, and no sample size was justified in advance by a detectable-effect calculation. Fourth, both substrates are short, self-contained Python functions with executable hidden tests, so nothing here measures auditing a change to a large program with no oracle, and both are drawn from benchmarks old enough that

memorisation cannot be ruled out. Fifth, the generator is a single model on both substrates, so the defect population is one model’s failure mode. Sixth, the residual rubric’s raters were a model and the author, who wrote the rubric and was not blind to the earlier labels; one external label set agrees with them across eleven overlapping tasks. Seventh, our attempt to build a population of specification-determined defects did not enforce the property, and its edits are separable from natural code at 96.7% [79.9, 97.3] by something the probe does not identify. Each of these is recorded in the study that carries it, together with the number it would change.

4.6. What we would do next

A severity-threshold sweep within one substrate, producing a recall–false-positive curve for each auditor family, is the experiment this paper most needs and the cheapest of these to run, because the findings are already archived and only the grading rule changes. It would decide whether the auditor contrast survives at a matched operating point. Replicating the cross-family arms with the sampling parameters matched would remove the confound above, and extending the cross-vendor family past eight readings would turn a statement about our budget into a statement about a curve. Then the experiment we would most like to have run: hold the code and the visible tests fixed and vary only the specification. For an instance that fails its hidden suite, add to the prose the behavioural rule it never settled — how ties break, what an empty input returns — without showing the failing case, and compare against a placebo edit of similar length that resolves nothing. If correct diagnosis rises under clarification and not under the placebo, the specification’s incompleteness is doing causal work; if it does not, our third result is a correlation between two properties of hard instances. That design tests the claim in a way neither the post-hoc re-rating nor the injected population can, because it manipulates the specification instead of conditioning on it and it does not disturb the code. Beyond that: adjudicating more than one reading of the shipped auditor would let a naming rate be compared with a union flag rate at the same K; a second generator would show whether the residual’s character is a property of the benchmark or of one model’s failures; a population of natural defects independently known to be specification-determined would separate the injected population’s artificiality from the property it was built to isolate; and a substrate whose ground truth is not a hidden test suite would test the assumption every ceiling here rests on, which is that a test suite asked the right question.

5. Related work

Critics that read code. Trained critic models write natural-language critiques of model-written code and outperform unaided humans at catching bugs (McAleese et al., 2024). That work evaluates on both deliberately inserted bugs and naturally occurring ones: on code containing naturally occurring model errors its critiques are preferred to human critiques in 63% of cases, and it surfaces errors in training data previously rated flawless. It also reports that critics hallucinate bugs, and that a human–model team hallucinates less than the model alone. So the natural-defect setting and the precision cost of a louder critic are both already its territory, and we do not claim either as new. And the reported quantity is a preference or a per-critique comparison rather than the union recall of repeated independent readings against an executed oracle. We should be plain that this work already occupies two of our four axes: it reports a precision–recall trade-off for critics, and it separates tampered from natural bugs. What we add on this axis is narrow — the union of repeated independent readings as an instrument for locating a limit, ground truth supplied by an interpreter rather than by preference, and the false-positive rate carried beside every recall figure. Aggregating several review passes is known to help: multi-review aggregation on 1,000 verified pull requests raises F1 by up to 43.67% (Zeng et al., 2025). That is the same mechanical move as our union-of- K , used as a method rather than as an instrument for finding a limit.

Repeated sampling and coverage. That coverage — the fraction of problems solved by at least one of K samples — rises steeply and then flattens is an established regularity of repeated sampling, reported over four orders of magnitude of K and fitted by a power law (Brown et al., 2024). Our union-of- K is that quantity with a flag rule in place of a correctness check, so §3.1 re-measures a known shape rather than discovering one, and at $K \leq 8$ our budget is small beside theirs. The contribution is not the shape. It is that the curve is read here as a measurement of a ceiling, reported beside the false-positive rate it costs, and tested against a registered flattening bar that it fails — which is why we report diminishing returns rather than a limit.

Self-assessment. That models are poor judges of their own output is established from two directions: self-correction without external feedback does not reliably improve reasoning and can degrade it (Huang et al., 2024), and models prefer their own generations when used as judges (Chen et al., 2025). Our same-vendor and self-audit arms are the code-audit case of that literature, and we read our null there as consistent with it rather than as new. What our arms add is the false-positive rate beside the recall. We also reported that the direction reverses on a second substrate; that claim is withdrawn with the study it came from (§3.5).

What the hidden suite decides. EvalPlus rebuilt HumanEval and MBPP with larger suites and reported defects in the original reference solutions (Liu et al., 2023); its code carries bespoke oracles for tasks where differential testing against the reference was not applicable. Richter & Papadakis (2026) show that models collapse onto a single incorrect reading of an underspecified task and name twelve MBPP tasks they identified by hand; eleven fall inside our defect population and our raters, who labelled them before that paper was fetched, agree on nine. From the test-generation side, the dominant error in generated tests is a wrong expected value rather than an invalid input (Xiong et al., 2023), and generated oracles tend to capture a program’s actual rather than its intended behaviour (Konstantinou et al., 2024). Our claim is the auditing counterpart: where the prose does not fix the value, a recall number computed against that suite measures the benchmark as much as the auditor.

Consensus as a substitute for ground truth. Several systems validate generated tests without a reference by agreement, through self-consistency, dual execution, or multi-agent discussion for JUnit suites (Taherkhani et al., 2026; Xu et al., 2025). In neither paper could we find the quantity that decides whether such a filter can be trusted: how many wrong assertions survive it. We measure it on a frozen set of canonical-failing tests and find that corroboration across independent generations does not reach our preregistered threshold, because wrong tests recur by problem (§3.5). This is the concrete case of a general result: self-consistent errors are exactly the class that every detection method they evaluate struggles to detect (Tan et al., 2025).

Who audits whom. That an evaluator should not be the model under evaluation is supported by measurements of self-preference in LLM judges against verifiable ground truth (Chen et al., 2025). Our contribution is its size against model-free ground truth, and the finding that its direction is substrate-dependent.

Preregistration. Preregistering an evaluation protocol is beginning to appear in machine learning, both as a protocol for experiments with agents (Vaccaro, 2026) and as a defence against tuning prompts, decoding parameters or output format until a desired result appears (Thomas et al., 2026). We record our use of it as method rather than contribution.

Limitations

Substrate. Every measurement here is on function-level Python tasks from two code benchmarks. Nothing in this paper establishes that the same ceiling, or the same explanation for it, holds for another language, for repository-scale changes, or for code whose correctness is not decided by an executable suite.

The ground truth is a test suite, and a test suite is not a specification. Hidden suites encode a reference implementation’s behaviour, including behaviour its prose never fixed. In the intervention study reported in §3.5, a generator shown an instance’s failing inputs inferred a behavioural rule that *contradicted* the hidden oracle on 23 of 32 instances. That is a property of the benchmarks, not of the generator, and it bounds what any recall figure measured this way can mean.

Intervals of unverified coverage. The primary intervals are percentile bootstraps over problem clusters whose coverage under this design has not been validated; in simulation the construction under-covers by roughly 2 to 5 points. Two earlier constructions in this programme were found by review to cover 0.416 and 0.075 where 0.95 was claimed. We state coverage as unverified rather than assume it.

The validation instrument shares the failure mode under study. Reports were checked by a language model, which is what this paper measures a ceiling on. Studies cleared review at rounds 4, 6, 7, 9, 10 and 11, so a single pass has low recall; no measurement ever moved, so on recomputation it is strong and on interpretation weak; and withholding our conclusions from one pass produced six substantive findings that six briefed passes had missed. The defect class it leaves open is the one that looks right to a language model, and we do not know what remains in it.

The reviewer is one of the measured families. `gpt-6-astra` reviewed every report in this programme and is also an auditor route whose ranking against the others the paper reports. For claims about `astra`’s relative performance the validation is a self-assessment, and those comparisons should be treated as bracketed until a model that audited nothing here has checked them.

No rater from outside the project. Every human judgement in this work is the first author’s. A third model rating does not discharge that, and the design that asks for an outside human has not been run.

Budget. Reading depth K , the number of arms, and the size of every intervention population were set by cost rather than by a power calculation, and no sample size was justified in advance by a minimum-detectable-effect argument.

Reproducibility and claims

The claims this paper is willing to defend, each with the interval and the label its source report gives it, are listed in `CLAIMS.md` in the repository at <https://github.com/dongzhaohe321418-lab/audit-ceiling-paper>, together with the claims the evidence does not support. Every preregistration, every numbered amendment, every report and every independent review is archived there verbatim, including the reviews

that refused approval. The repository also carries the auditor’s constitution, the exact prompts for the auditor, the generator and each rating rubric, the capability card of every route with its sampling parameters, the frozen instance manifests with their hashes, and the analysis code that produces every number and figure below from those records. What the repository cannot carry is the raw run directories themselves. They contain model outputs derived from corpora under a non-commercial, no-redistribution licence, so what is published is every derived record – scores, counts, hashes, labels, prompts and manifests – and not the bytes they were derived from. A replication therefore reproduces every number in this paper from the committed records, and reproduces the records themselves only by re-running the studies against the same corpora. The repository is private while the remaining study is in review and is released with the paper.

References

- Brown, B., Juravsky, J., Ehrlich, R., Clark, R., Le, Q. V., Ré, C., and Mirhoseini, A. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*, 2024.
- Chen, W.-L., Wei, Z., Zhu, X., Feng, S., and Meng, Y. Do LLM evaluators prefer themselves for a reason? *arXiv preprint arXiv:2504.03846*, 2025.
- Huang, J., Chen, X., Mishra, S., Zheng, H. S., Yu, A. W., Song, X., and Zhou, D. Large language models cannot self-correct reasoning yet. *arXiv preprint arXiv:2310.01798*, 2024.
- Konstantinou, M., Degiovanni, R., and Papadakis, M. Do LLMs generate test oracles that capture the actual or the expected program behaviour? *arXiv preprint arXiv:2410.21136*, 2024.
- Liu, J., Xia, C. S., Wang, Y., and Zhang, L. Is your code generated by ChatGPT really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2305.01210.
- McAleese, N., Pokorny, R. M., Uribe, J. F. C., Nitishinskaya, E., Trebacz, M., and Leike, J. LLM critics help catch LLM bugs. *arXiv preprint arXiv:2407.00215*, 2024.
- Richter, C. and Papadakis, M. Underspecification does not imply incoherence: The risks of semantic collapse in coding models. *arXiv preprint arXiv:2607.01953*, 2026.
- Taherkhani, H., DaghighFarsoodeh, A., Chowdhury, M., Pham, H. V., and Hemmati, H. Consistency meets verification: Enhancing test generation quality in large lan-

guage models without ground-truth solutions. *arXiv preprint arXiv:2602.10522*, 2026.

Tan, H., Sun, F., Liu, S., Su, D., Cao, Q., Chen, X., Wang, J., Cai, X., Wang, Y., Shen, H., and Cheng, X. Too consistent to detect: A study of self-consistent errors in LLMs. *arXiv preprint arXiv:2505.17656*, 2025.

Thomas, M., Gligorić, K., and Shah, N. B. Mitigating LLM-based p-hacking by preregistering for the next LLM. *arXiv preprint arXiv:2606.27687*, 2026.

Vaccaro, M. Preregistration for experiments with AI agents. *arXiv preprint arXiv:2606.11217*, 2026.

Xiong, W., Guo, Y., and Chen, H. The program testing ability of large language models for code. *arXiv preprint arXiv:2310.05727*, 2023.

Xu, Q., Wang, G., Briand, L., and Liu, K. Hallucination to consensus: Multi-agent LLMs for end-to-end JUnit test generation. *arXiv preprint arXiv:2506.02943*, 2025.

Zeng, Z., Shi, R., Han, K., Li, Y., Sun, K., Wang, Y., Yu, Z., Xie, R., Ye, W., and Zhang, S. SWR-Bench: Assessing LLM performance in real-world code review comment generation. *arXiv preprint arXiv:2509.01494*, 2025.